



Microsoft 365 Telemetry Cheat Sheet

Where M365 audit data actually comes from — configuration surfaces, log tables, collection paths, and the blind spots hiding at every layer.

1. THE FIVE CONFIGURATION SURFACES

There is no unified view of what is logged across an M365 tenant. Each surface has its own portal, RBAC, and failure mode. None know about the others.

#	SURFACE	WHERE	CONTROLS	COMMON GAP
1	Unified Audit Log (UAL)	Purview Compliance Portal	Tenant-level audit master switch	OFF by default for SMB SKUs (Business Basic / Standard / Premium) and unmanaged enterprise trials; ON by default for E3 / E5 / G3 / G5
2	Mailbox Audit Config	Exchange Online — PowerShell only	Per-mailbox actions logged for Owner / Delegate / Admin	Customized mailboxes fall out of DefaultAuditSet ; new actions never auto-added
3	Entra ID Diagnostic Settings	Entra portal → Monitoring	Sign-in / audit / provisioning / risk / Graph API logs → LA workspace, Event Hub, or Storage	Never configured = no sign-in logs, no MFA detail, no conditional access context in SIEM
4	MDCA M365 Connector	Defender XDR → Settings → Cloud apps → App connectors	Populates CloudAppEvents	"Microsoft 365 activities" checkbox unselected → table empty silently; no error; no alert
5	Mgmt Activity API Subscriptions	Programmatic — per content type	Pulls audit blobs into SIEM (Audit.Exchange, .SharePoint, .AzureActiveDirectory, .General, DLP.All)	Audit.General not subscribed → Teams, Power Platform, Copilot, and Defender XDR audit invisible

Most-missed:

Audit.General. Most teams subscribe to Exchange, SharePoint, AzureActiveDirectory and stop there. There is no health monitoring, no notification when a subscription fails, and no completeness check across the five surfaces.

2. LOG TABLES AT A GLANCE

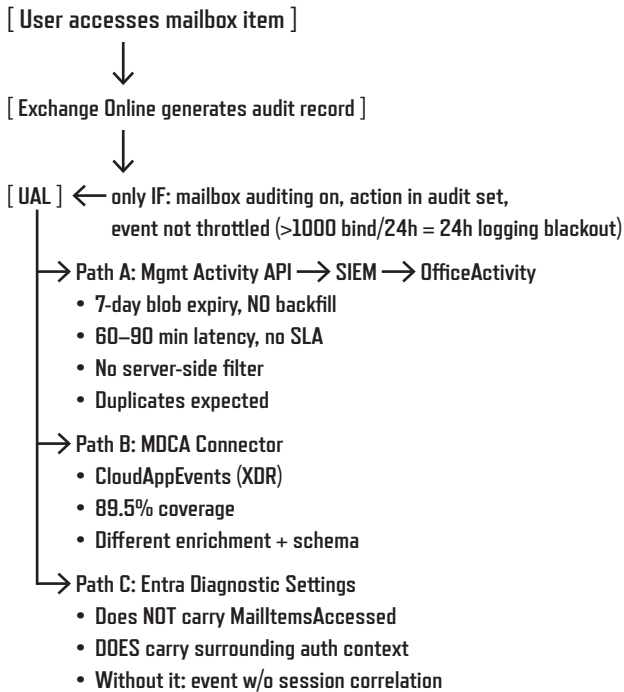
TABLE	PLATFORM	SOURCE	LICENSE	RETENTION	COVERAGE / NOTES
OfficeActivity	Sentinel	Mgmt Activity API	M365 connector (free)	90d free w/ Sentinel	~40% — missing OperationCount
CloudAppEvents	Defender XDR	MDCA connector	E5 Security or MDCA standalone	30d (extendable via Sentinel Data Lake)	89.5% — no UserLoggedIn / UserLoginFailed
Purview Audit Search	Purview	Direct UAL query	E3 Standard / E5 Premium	180d / 1yr / up to 10yr add-on	~99.5% — IR canonical source, not a SIEM table
EntraIdSignInEvents	Defender XDR	Entra sign-in logs	Entra ID P2	30d	Interactive + non-interactive user auth (GA Feb 2026)
EntraIdSpnSignInEvents	Defender XDR	Entra SPN / MI sign-ins	Entra ID P2	30d	Service principal + managed identity auth (GA Feb 2026)
GraphApiAuditEvents	Defender XDR	Auto with Defender XDR	Included with XDR	30d in XDR; streamable to Sentinel	Graph API — 20 cols ; missing DeviceId / SessionId
MicrosoftGraphActivityLogs	Sentinel	Entra Diagnostic Settings	Entra P1 + Sentinel	Workspace-configured	Graph API — full 35 cols ; expensive ingest
SignInLogs	Sentinel	Entra Diagnostic Settings	Entra ID P1	Workspace-configured	Interactive auth — richest auth schema (CA, MFA, device, risk)
AADNonInteractiveUserSignInLogs	Sentinel	Entra Diagnostic Settings	Entra ID P1	Workspace-configured	Token refresh / SSO — high volume, often overlooked
AADServicePrincipalSignInLogs	Sentinel	Entra Diagnostic Settings	Entra ID P1	Workspace-configured	SPN auth — Sentinel equivalent of EntraIdSpnSignInEvents
AADManagedIdentitySignInLogs	Sentinel	Entra Diagnostic Settings	Entra ID P1	Workspace-configured	Managed identity auth
AuditLogs	Sentinel	Entra Diagnostic Settings	Entra Free / E3+	Workspace-configured	Directory changes — only table here w/o premium gate

Schema gotchas:

- OfficeActivity missing **OperationCount** → cannot distinguish aggregated from single events. >1/3 of events are aggregated. Counts and thresholds can be inaccurate, with no indication in the data itself.
- GraphApiAuditEvents missing **DeviceId** / **SessionId**; **UserId** + **ServicePrincipalId** concatenated into **AccountObjectId**.
- CloudAppEvents UAL. Some events (e.g., OneDrive **Broke sharing inheritance**) appear only in **CloudAppEvents**. The relationship is overlapping, not hierarchical.
- Diagnostic settings naming trap: category names ≠ table names. **NonInteractiveUserSignInLogs** → **AADNonInteractiveUserSignInLogs**. Same for **ServicePrincipalSignInLogs** / **ManagedIdentitySignInLogs**.
- Coverage percentages are based on Bert-Jan Pals' testing of 191 unique activities in default-configured tenant.

3. COLLECTION PATHS

Trace of `MailItemsAccessed` — the Storm-0558 detection event:



Critical Insight:

- Your SIEM does not query the UAL. It queries whatever subset made it through one specific pipe. Different collection paths result in different coverage, latency, retention, and schema — even for the “same” event.

4. VERIFICATION – POWERSHELL

Tenant-level UAL ingestion — must be run in Exchange Online PowerShell; Security and Compliance PS returns false negative.

```
Get-AdminAuditLogConfig |  
Select-Object UnifiedAuditLogIngestionEnabled
```

Per-mailbox audit config — the only reliable verification method; no portal UI exists.

```
Get-Mailbox -Identity user@domain.com |  
Select-Object AuditEnabled, AuditOwner, AuditDelegate, AuditAdmin,  
DefaultAuditSet
```

Bulk verification — find drifted mailboxes

```
Get-Mailbox -ResultSize Unlimited |  
Where-Object { $_.AuditEnabled -eq $false -or  
$_DefaultAuditSet -notcontains 'Owner' } |  
Select UserPrincipalName, AuditEnabled, DefaultAuditSet
```

Mgmt Activity API subscription status

```
# Requires app registration + token  
GET https://manage.office.com/api/v1.0/  
{tenantId}/activity/feed/subscriptions/list
```

REFERENCES & CREDITS

Bert-Jan Pals — UAL = Unaligned Activity Logs (kqlquery.com, Nov 2024); GraphApiAuditEvents (kqlquery.com, Aug 2025).
CISA — Microsoft Expanded Cloud Logs Implementation Playbook (Jan 2025).
Fabian Bader (Cloudbrothers) — Detect threats using GraphApiAuditEvents — Part 3 (Aug 2025).
Jeffrey Appel — 2025 Microsoft Defender Optimization & Configuration Cheat Sheet (Nov 2025).

5. BLIND SPOT CHECKLIST

CONFIGURATION

- ☐ UAL on, but mailbox actions not in DefaultAuditSet — one historical customization = permanent drift
- ☐ MDCA connector “connected” but M365 Activities checkbox not selected → CloudAppEvents empty silently
- ☐ Mgmt Activity API missing Audit.General → Teams, Power Platform, Copilot, Defender XDR audit invisible
- ☐ Entra ID Diagnostic Settings not configured → no sign-in logs in SIEM
- ☐ Event Hub destination missing for non-Microsoft SIEM → identity blind on Splunk / Elastic / etc.

COVERAGE

- ☐ Detections built only against OfficeActivity → may miss approximately 60% of activity
- ☐ Detections rely on event counts → missing OperationCount can lead to inaccurate thresholds (1/3 of events aggregated)
- ☐ Auth detections only on EntraIdSignInEvents → blind to SPN / MI auth (different table)
- ☐ Graph API detections need DeviceId / SessionId → GraphApiAuditEvents won't deliver
- ☐ Assumed CloudAppEvents / UAL → false; relationship is overlapping, not hierarchical

RETENTION

- ☐ Mgmt Activity API blobs expire at 7 days → ingestion outage = permanent gap, no backfill
- ☐ GraphApiAuditEvents 30d in XDR; Sentinel forwarding via Defender XDR connector — verify it's actually configured for IR
- ☐ CloudAppEvents 30d in XDR — Data Lake ingestion not configured
- ☐ SIEM retention is irrelevant for events that were never collected

The Meta Blind Spot:

No native health monitoring exists for any of this. No alert when an API subscription fails. No “what’s flowing” dashboard. No completeness check. “Nothing happened” looks identical to “we stopped collecting.”

6. WHAT TO DO

- 1. Identify your specific ingestion path.** Don't guess. Confirm Mgmt Activity API vs. MDCA vs. Entra Diagnostic Settings vs. combination. Know which tables your detections query.
- 2. Verify the Audit.General subscription.** Without it, Teams, Power Platform, Copilot, and Defender audit activity remain invisible.
- 3. Verify per-mailbox audit config with PowerShell** — especially executives, finance, legal, IT admins, and known targets.
- 4. For IR, supplement SIEM data with direct UAL extraction.** Invictus Microsoft Extractor Suite, Purview Audit Search, or PowerShell. SIEMdata reflects what made it through the collection pipeline; the UAL reflects what actually happened.
- 5. Cross-check coverage.** Don't assume OfficeActivity = UAL. Don't assume CloudAppEvents = UAL. They are different pipes with overlapping but non-hierarchical coverage.
- 6. Add GraphApiAuditEvents as a baseline for Graph API visibility.** GA Feb 2026, free with Defender XDR. Thinner schema than MicrosoftGraphActivityLogs but better than zero.
- 7. Build your own health monitoring.** Microsoft does not provide it. Continuous verification that data is flowing is the only defense against the meta blind spot.

Thomas Naunheim — Sign-in logs and auditing of Managed Identities and Service Principals (cloud-architekt.net).

Microsoft Learn — Mgmt Activity API reference; Audit mailboxes; Entra Diagnostic Settings; Defender XDR What's New.

CISA AA23-193A — Storm-0558 advisory.

Companion blog post: *Where Your M365 Telemetry Actually Comes From* — lydiagraslie.substack.com